The transfer function of subset-sum landscapes: rigidity of the gap series off centre

Kentaro Amauchi*

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Abstract

For a finite set A of odd integers, the number of strict local minima of the subset-sum landscape at a target n, divided by the number of ground states, is governed at the centre $n = \lfloor T/2 \rfloor$ by the gap series $\Gamma(A) = 1 + 2\sum_{d\geq 1} 2^{-\#\{a\in A:\, a\leq 2d\}}$. This paper argues that this is a property of the landscape rather than of its centre, and in a sharper form: that the ratio is a universal functional of a single local observable, the logarithmic slope λ of the representation count at the target, through an explicit transfer function $\Phi(\lambda) = 1 + \sum_{d\geq 1} 2^{1-N_d} e^{-\lambda^2 s_d/8} \cosh(\lambda\delta_d)$ built from the small elements of A alone; rigidity is the case $\lambda \to 0$. We prove the analytic ingredients — an additive bridge carrying the minor-arc estimates of Part II [2] from the uniform measure to the exponential tilt, the minor-arc input in the form that bridge needs, and the threshold $\alpha = 1$ below which the governing hypothesis fails — and assemble them into a proof skeleton for the two theorems, the remaining gap being the Edgeworth constants of a classical local limit computation. The extremal denominator of the bridge is $q = 6$, the same one that produces the constant $\sqrt{3}/2$ in the companion paper. The transfer function is verified against exact computation on four profiles: the residual is 0.8–1.9% of the signal and obeys a c_A/k law whose constant is predicted, with no free parameter, by the profile alone.

1 Introduction

Parts I and II [1, 2] of this series establish that for the odd primes $\mathrm{lm}/\deg \to \Gamma$ *at the centre* of the landscape. Whether Γ is an invariant of the landscape or an accident of its most symmetric point is the question this paper answers, and the answer turns out to be stronger than rigidity alone: the whole ratio is a function of one local observable. The argument and what each part rests on:

- Parts I and II establish $\mathrm{lm}/\deg \to \Gamma$ *at the centre* $n = \lfloor T/2 \rfloor$: Part I reduces it to a flatness hypothesis and Part II discharges that hypothesis unconditionally. The obvious question — whether Γ is a property of the landscape or of its centre — is what this paper answers.
- The combinatorial layer (classification, exact stratification, sandwich) was proved *for every target* in Part I and is formally verified; nothing there needs redoing. [Lean-verified]
- The minor-arc estimates of Part II [2] bound $|G_B(\theta)|$ as a function of θ only. They do not refer to the extraction point m, which enters $r_B(m) = \frac{1}{2\pi}\int F_B(\theta)e^{-im\theta}d\theta$ only through a phase. They therefore transfer to off-centre targets unchanged. [quoted from Part II]

*Independent researcher.

- What is new is the middle, §4.2: an additive bridge (Lemma 4.8) carrying the minor-arc estimates from the uniform measure to the exponential tilt, the minor-arc input in the form the bridge needs (Lemma 4.10), and a tilted local limit theorem uniform over the extraction window (Proposition 4.20). The odd orders in λ cancel by the reflection symmetry of the landscape, which is where cosh enters.
- One consequence of quoting rather than reproving deserves saying at the outset: *the only ineffective constant in the trilogy is one and the same Siegel–Walfisz constant, and it enters each paper at exactly one point* — here, part (b) of Lemma 4.10. See §7.

2 Notation, and what is inherited

The definitions are those of Part I [1], written so that nothing assumes $n = T/2$. Fix throughout an increasing sequence $A = (a_1, \dots, a_k)$ of odd positive integers, $T = \sigma(A)$, $S_2 = \sum a_i^2$, $V = S_2/4$, and a target n. Write $\rho = n/T$ and $x = \frac{1}{2} - \rho$. For $d \geq 1$ put

$$N_d = \#\{i : a_i \leq 2d\}, \qquad \sigma_d = \sum_{a_i \leq 2d} a_i, \qquad s_d = \sum_{a_i \leq 2d} a_i^2, \qquad \delta_d = d + \tfrac{\sigma_d}{2},$$

and let $B_d = \{a_i : a_i > 2d\}$ be the layer-d ensemble, of variance $V_d = (S_2 - s_d)/4 = V - s_d/4$.

Definition 2.1 (local logarithmic slope)**.** $\lambda(n)$ is the logarithmic derivative of the representation count at the target,

$$\lambda(n) \;=\; +\frac{d}{dm} \log r_A(m)\Big|_{m=n} \;=\; \tfrac{1}{4} \log \frac{r_A(n+2)}{r_A(n-2)} + O(\cdot),$$

so $\lambda > 0$ below the centre, where r_A is increasing.

Remark 2.2 (on the sign, and why it took until paper 4 to matter). An earlier version of this definition carried the opposite sign, which made it disagree with Remark 2.3 (where $\lambda \approx (T/2 - n)/V > 0$ below the centre) and with Remark 2.5 (where $s/\lambda \to 1$ with $s > 0$ below the centre). Nothing in this paper changed when the sign was corrected, and nothing could have: Φ is an *even* function of λ, so the sign of λ is not observable anywhere in the transfer function. It becomes observable only when the cosh is split into two unequal exponentials, which is what happens under a biased measure; the discrepancy was found there and is recorded here. The convention above is the one that makes $\lambda \approx s$.

Remark 2.3 (why the slope and not the offset). The Gaussian approximation $\lambda \approx (T/2 - n)/V$ is what one writes down first, and it is wrong at the 10% level for the purposes of this paper. Substituting the measured λ for the Gaussian one takes the drift of the fitted coefficient across the range $x \in [0.06, 0.30]$ from 15–18% down to 0.03–0.07%, and it dissolves a non-integer exponent that an earlier draft of this programme had reported as a finding: the exponent was an artefact of the wrong variable. This is not a cosmetic choice: every statement below is false at the stated precision if $(T/2 - n)/V$ is substituted for λ.

Definition 2.4 (the tilt)**.** P_s is the product measure on subsets under which a_i is present with probability $p_a = (1 + e^{sa})^{-1}$, and $s = s(n) > 0$ is chosen so that $\sum_a a\, p_a = n$; thus P_s is the exponential tilt of the uniform measure whose mean sits at the target.

Remark 2.5 (the tilt and the slope agree to first order). s and λ are the same quantity to leading order but not exactly: s makes the tilted *mean* equal n, whereas λ is the slope of $\log r_A$ *at* n, and the two differ by the mean–mode gap. Measured on three profiles,

$$\frac{s(n)}{\lambda(n)} \;=\; 1 + \frac{c_A}{k} + O(k^{-2}), \qquad c_A \approx 1.85 \text{ (odds)},\ 2.03 \text{ (primes)},\ 2.87 \text{ (squares)},$$

stable in x and clean in k over $k = 100, \ldots, 220$. [experimentally confirmed; `tilt_r088`] The statements below are formulated in λ, and *not* in s: substituting s multiplies the residual of Table 1 by three, exactly as the λ^2 leading behaviour of Φ predicts. [experimentally confirmed; `sres_r088`]

3 Rigidity

Theorem 3.1 (rigidity of the gap series). *[proof skeleton, and the missing ingredient is named rather than left to be located. The analytic ingredients are proved in §4.2; what is* not *written out is the Edgeworth expansion in region R1 of Proposition 4.20 — a classical local-limit computation under a product measure, given here only to the level of its cumulant budget, with the explicit constants missing. Until those constants are written, the statement below is conditional on that computation and on nothing else.] Let $P = P_k$ be the first k odd primes and let $\rho \in (0,1)$ be fixed. Then*

$$\frac{\mathrm{lm}_P(\lfloor \rho T \rfloor)}{\deg_P(\lfloor \rho T \rfloor)} \xrightarrow[k\to\infty]{} \Gamma(P),$$

uniformly for ρ in compact subsets of $(0,1)$. For a general profile A the same holds under hypothesis (H) *of §6.*

$\Gamma(A)$ is therefore an invariant of the landscape and not of its centre. Theorem 3.1 is the case $\lambda \to 0$ of Theorem 4.1: at fixed ρ one has $\lambda(n) \asymp x/N \to 0$, and Φ is continuous at 0 with $\Phi(0) = \Gamma$, so

$$\Phi(\lambda) - \Gamma \;=\; O\Big(\lambda^2 \sum_{d\ge 1} 2^{1-N_d}\delta_d^2\Big) \;\longrightarrow\; 0,$$

the sum being finite precisely by (H). The uniformity in ρ needs no new estimate: every bound in §4.2 is uniform once $t_{\min}(\rho)$ is bounded below, and $t_{\min}$ is continuous and positive on $(0,1)$, hence bounded below on compact subsets (Remark 4.19).

[experimentally confirmed] The evidence the theorem is modelled on. With $\mathrm{ratio}(\rho, k) = [\mathrm{lm}/\deg]/\Gamma(P_k)$, the spread over ρ contracts monotonically:

k	40	52	64	76	88	100
$\max_\rho - \min_\rho$	0.01051	0.00283	0.00257	0.00162	0.00113	0.00083
$\max_\rho \lvert\mathrm{ratio} - 1\rvert$	0.00921	0.00276	0.00244	0.00156	0.00109	0.00081

over $\rho \in [0.20, 0.50]$, by exact integer computation. Three guards are in force throughout this paper's computations and are worth stating once: every reported point satisfies $\deg \ge 10^4$, so that integer granularity cannot masquerade as signal; the dynamic programme was checked against brute-force enumeration over all 2^k subsets at small k; and the exact identity $\mathrm{lm}(n) = \mathrm{lm}(T - n)$ is verified at every measured target, which tests the stratification and the arithmetic together.

4 The transfer function

Theorem 4.1 (transfer). *[proof skeleton, and the missing ingredient is named rather than left to be located. The analytic ingredients are proved in §4.2; what is* not *written out is the Edgeworth expansion in region R1 of Proposition 4.20 — a classical local-limit computation under a product measure, given here only to the level of its cumulant budget, with the explicit constants missing. Until those constants are written, the statement below is conditional on that computation and on* nothing else.] *Let $A = P_k$ be the first k odd primes, or a general profile satisfying* (H)*, and let $n = \lfloor \rho T \rfloor$ with $\rho \in (0,1)$ fixed. Then*

$$\frac{\mathrm{lm}_A(n)}{\deg_A(n)} = \Phi_k(\lambda(n)) + E_k, \qquad \Phi_k(\lambda) = 1 + \sum_{d \geq 1} 2^{1-N_d} e^{-\lambda^2 s_d/8} \cosh(\lambda \delta_d), \tag{1}$$

with $E_k \to 0$ as $k \to \infty$, uniformly for ρ in compact subsets of $(0,1)$.

Remark 4.2 (what the theorem does and does not claim about E_k). The assertion is $E_k \to 0$ in absolute terms, nothing sharper. The measured rate $E_k \asymp c_A/k$ of Remark 4.7 is *not* part of the statement; it is quoted as [experimentally confirmed, $k \leq 220$, three profiles] and stated as an open problem in §7. Distinguishing the two matters here more than usual, because the numerical agreement is far better than what is proved.

Three features of (1) deserve to be stated before any proof.

Remark 4.3 (the ratio sees the target only through one number). Φ depends on A alone. The target enters only through $\lambda(n)$. This is a stronger statement than rigidity: rigidity says the limit does not move, (1) says how it fails to move.

Remark 4.4 ($\lambda = 0$ recovers Part I [1]). The layer weight $e^{-\lambda^2 s_d/8}$ is 1 at $\lambda = 0$, so $\Phi(0) = 1 + \sum_{d \geq 1} 2^{1-N_d}$, which is the window identity $W_D(A) = \Gamma(A) + (2D+1)2^{-k}$ of Part I — already formally verified. Numerically, $\Phi(0) - \Gamma(A)$ is 0, $-4.7 \cdot 10^{-14}$ and $-2.7 \cdot 10^{-15}$ for the three sequences of Table 1.

Remark 4.5 (each layer is a smaller ensemble). The factor $e^{-\lambda^2 s_d/8}$ is not a fudge: it is what remains when the Gaussian normalisations of B_d and of A fail to cancel, because $V_d = V - s_d/4$. Expanding to second order gives the coefficient of λ^2 as $\delta_d^2 - s_d/4$ rather than δ_d^2. In the notation of §2, layer d is the ensemble B_d with the small elements removed, so its variance is $V_d = V - s_d/4$ rather than V; the weight $e^{-\lambda^2 s_d/8}$ is exactly the price of that difference, and Remark 4.21 derives it from the ratio of partition functions rather than fitting it. The correction is termwise smaller than the term it corrects, $s_d/4 \leq d\sigma_d/2 < \delta_d^2$, so every term of the series stays positive and the correction never overshoots.

4.1 Verification

[experimentally confirmed] Table 1 evaluates (1) at the measured $\lambda(n)$ against the exactly computed $\mathrm{lm}/\deg$.

$x = \frac{1}{2} - \rho$	0.06	0.08	0.10	0.15	0.20	0.25	0.30
odds, $k = 220$	0.84%	0.85%	0.85%	0.86%	0.88%	0.92%	0.97%
primes, $k = 220$	0.91%	0.92%	0.92%	0.94%	0.96%	0.99%	1.05%
$a_i \asymp i^{3/2}$, $k = 220$	1.05%	1.06%	1.06%	1.08%	1.10%	1.14%	1.20%
squares, $k = 170$	1.66%	1.67%	1.68%	1.70%	1.74%	1.80%	1.89%

Table 1: Residual $|\text{measured} - \Phi(\lambda)|$ **as a fraction of the centred signal** $\mathrm{lm}/\deg - [\mathrm{lm}/\deg]_{n=T/2}$. Absolute residuals are 10^{-5} to 10^{-6}; quoting them as significant figures of the ratio would be meaningless, since the signal itself lives in the fifth and sixth digit. The third row is the only ensemble in this paper that was *constructed to test a prediction* rather than chosen because it was natural: $\alpha = 3/2$ was built after Remark 4.7 had been fitted on the other three, and its predicted constant 16/7 lies between theirs.

Remark 4.6 (why the residual is quoted this way). Against the *uncentred* signal the same residuals read 2% to 300%, the large values occurring precisely where the signal is smallest.

Neither number is wrong; the centred one is the one the theorem is about, because $\Phi(0) = \Gamma$ is an identity and the finite-k offset $[\mathrm{lm}/\deg]_{n=T/2} - \Gamma$ is a separate quantity of size 10^{-6}. We keep this remark because an earlier draft of this programme reported "agreement to five or six significant figures" for a model that was 12% wrong on the signal: the figures were real and the claim was not, because the signal itself lives in those digits.

Remark 4.7 (the residual obeys a $1/k$ law). Table 1 is one column of a k-scan. Writing $R_A(k)$ for the residual in the sense of the caption, the measurement over $k = 60, \ldots, 220$ gives

$$k\,R_A(k) \;\longrightarrow\; \begin{cases} 1.85\% & \text{odds } (k = 60, 100, 140, 180, 220:\;\; 2.29, 2.00, 1.91, 1.87, 1.85), \\ 2.01\% & \text{primes } (k = 100, \ldots, 220:\;\; 2.38, 2.06, 2.03, 2.01), \\ 2.82\% & \text{squares } (k = 130, \ldots, 190:\;\; 3.78, 2.85, 2.83, 2.82). \end{cases}$$

Two things follow. First, the residual is $O(1/k)$ and therefore vanishes, which is the form Theorem 4.1 asserts; the table's 1% is a finite-k number and not a floor. Second, the squares row is the loosest of the three because c_A is larger, not because the decay is different: squares obey the same law with a constant 1.52 times that of the odd numbers. The constants c_A coincide numerically with those of Remark 2.5 — the same mean–mode gap seen twice — but substituting s for λ does not remove the residual, so the coincidence is a shared order of magnitude and not a shared cause. [experimentally confirmed; `tilt_r088`, `sres_r088`]

What they share is a driver. The saddle-point representation gives the exact identity

$$s(n) - |\lambda(n)| \;=\; \frac{K'''(s)}{2K''(s)^2}, \qquad K'' = \sum a^2 p_a(1-p_a), \quad K''' = \sum a^3 p_a(1-p_a)(1-2p_a),$$

tested pointwise to a relative error $0.54/k$ [experimentally confirmed; `saddle_r092`], and for small tilt $K'''/(2K''^2) \approx sS_4/S_2^2$, so that $c'_A = k(s/|\lambda| - 1) \to kS_4/S_2^2$. For $a_i \asymp i^\alpha$ this is $(2\alpha+1)^2/(4\alpha+1)$. The measured c_A of this remark agrees with c'_A to within 1% on four profiles:

	α	kS_4/S_2^2	c'_A	c_A	c_A/c'_A
odd numbers	1	1.8000	1.8525	1.8674	1.008
$a_i \asymp i^{3/2}$	3/2	2.2794	2.3494	2.3318	0.993
squares	2	2.7714	2.8615	2.8377	0.992
primes	—	1.9731	2.0251	2.0253	1.000

at $k = 220$. The two halves of this have different statuses and the difference matters:

- $c'_A \to (2\alpha+1)^2/(4\alpha+1)$ is *derived* — the identity above is exact, its sign is forced by $\Lambda''' < 0$ for $\rho < \frac{1}{2}$, and the small-tilt step is a Taylor expansion — and [experimentally confirmed on five profiles, one of them out of sample].

- $c_A = c'_A(1 + o(1))$ is [experimentally confirmed: four profiles, ratios 0.992–1.008, $k = 220$]. The exact asymptotic equality, and the mechanism behind it, remain a **conjecture**. What is established is a shared driver, S_4/S_2^2, and *not* a causal link: substituting s for λ makes the residual three times worse, not smaller.

Carried to $k = 300$, the identification tightens rather than drifting, and it does so at a measurable rate. The ratio c_A/c'_A reads

k	120	180	220	260	300
odd numbers	1.0589	1.0244	1.0151	1.0098	1.0064
primes	1.0318	1.0084	1.0039	1.0014	1.0001
$a_i \asymp i^{3/2}$	1.0113	1.0015	0.9995	0.9984	—
squares	1.0782	1.0039	0.9985	—	—

so $|c_A/c'_A - 1|$ falls by a factor 9.2 across a factor 2.5 in k for the odd numbers, an exponent of 2.4 — the gap between the two constants closes *faster* than the $1/k$ law they both sit on. [experimentally confirmed, four profiles, $k \le 300$; e8_r103] This is a sharper statement of the same conjecture, not a proof of it: a rate of empirical convergence is evidence about a limit, and the mechanism is still missing.

4.2 The bridge to Part II

Both theorems reduce to the same five parts. **P1** is the exact combinatorial stratification, proved for every target in Part I [Lean-verified]; **P3** is a local limit theorem for the tilted ensemble of Definition 2.4; **P4** is the assembly, in which the odd orders in λ cancel by the reflection symmetry $\operatorname{lm}(n) = \operatorname{lm}(T-n)$ [Lean-verified]; **P5** is the error budget, which converges exactly under (H). Only **P2** is new: the minor-arc estimates of Part II [2] are proved for the *uniform* measure, and P3 needs them for the tilted one.

Write $\widetilde{G}_B(\theta) = \prod_{a\in B}(1 - p_a + p_a e(a\theta))$ for the tilted generating function and $S_B(\theta) = \sum_{a\in B} e(a\theta)$ for the linear exponential sum.

Lemma 4.8 (the bridge)**.** *Put $t_a = 4p_a(1-p_a)$ and $t_{\min} = \min_{a\in B} t_a$. For every θ,*

$$|\widetilde{G}_B(\theta)|^2 = \prod_{a\in B}\Big(1 - t_a\sin^2(\pi a\theta)\Big) \;\le\; \exp\Big(-t_{\min}\sum_{a\in B}\sin^2(\pi a\theta)\Big) \;=\; \exp\Big(-\tfrac{t_{\min}}{2}\big(b - \operatorname{Re} S_B(\theta)\big)\Big).$$

Consequently, if $\operatorname{Re} S_B(\theta) \le \eta\, b$ then $|\widetilde{G}_B(\theta)| \le \exp(-t_{\min}(1-\eta)b/4)$.

Proof. The identity $|1 - p + pe(\phi)|^2 = 1 - 4p(1-p)\sin^2(\pi\phi)$ is a computation; $t_a \ge t_{\min}$ and $1 - u \le e^{-u}$ give the middle step; and $\sum_a \sin^2(\pi a\theta) = \frac{1}{2}(b - \operatorname{Re} S_B(\theta))$ is the double-angle formula summed. □

Remark 4.9 (why the bound is additive, and a bound that is not available). It is tempting to transfer multiplicatively, as $|\widetilde{G}_B(\theta)| \le |G_B(\theta)|^\kappa$ with G_B the untilted product, since that would carry every exponent of Part II across at once. **No such $\kappa > 0$ exists.** At any θ with $a\theta \in \frac{1}{2} + \mathbb{Z}$ for some $a \in B$ the untilted factor $\cos(\pi a\theta)$ vanishes, so the right-hand side is 0, while the tilted factor equals $\sqrt{1 - t_a} > 0$. Equivalently $\log(1 - ty)/\log(1-y) \to 0$ as $y \to 1$. The underlying inequality $1 - ty \le (1-y)^t$ is false for every $t \in (0,1)$ — $s \mapsto (1-y)^s$ is convex and therefore lies *below* its chord $1 - sy$, with deficit exactly $-(1-t)$ at $y = 1$. Lemma 4.8 avoids the issue by never dividing by a quantity that vanishes. [refuted numerically on a 40 575-point grid and structurally; kappa_r088]

Lemma 4.10 (the minor-arc input)**.** *[(a) proved, Lemma 4.11; (b) proved, Lemma 4.12; (c) quoted from Part II] Let $A = P_k$ and $N = a_k$. Then*

$$\operatorname{Re} S_A(\theta) \;\le\; \Big(\tfrac{1}{2} + o(1)\Big)b \qquad \textit{uniformly for } \tfrac{1}{2N} \le \theta \le 1 - \tfrac{1}{2N}.$$

(a) $q \in \{1, 2\}$*. Lemma 4.11 below.*

(b) $3 \le q \le (\log N)^{A_0}$*. Lemma 4.12 below.*

(c) $q > (\log N)^{A_0}$. This is the case $n=1$ of Step 2 of `prop:deepminor` *in Part II, which gives $|S_A(\theta)| \ll N(\log N)^{3-A_0/2} + N^{4/5}(\log N)^3$, i.e. $\operatorname{Re} S_A/b \ll (\log N)^{4-A_0/2} = o(1)$ for $A_0 > 8$. [quoted from Part II, verbatim]*

Lemma 4.11 (the two degenerate denominators)**.** *Let $q \in \{1,2\}$, $\theta = a/q+\beta$ with $\|\beta\| \geq 1/(2N)$. Then $\operatorname{Re} S_A(\theta) \leq \frac{1}{2}b$ for all large N.*

Proof. Write $S_A(\theta) = \sum_{p\leq N} e(pa/q)e(p\beta)$. For $q=1$ the first factor is 1; for $q=2$ it is $(-1)^p = -1$ on every odd prime, so $S_A = -\sum_{p\leq N} e(p\beta)$ and it suffices in both cases to bound $|\sum_{p\leq N} e(p\beta)|$. Partial summation against $\pi(t) = \operatorname{li}(t) + O(te^{-c\sqrt{\log t}})$ gives

$$\sum_{p\leq N} e(p\beta) = \int_2^N e(t\beta)\frac{dt}{\log t} + O\big(Ne^{-c\sqrt{\log N}}(1+|\beta|N)\big),$$

and integrating by parts once more, $|\int_2^N e(t\beta)\,dt/\log t| \ll 1/(|\beta|\log N)$. With $\|\beta\| \geq 1/(2N)$ this is $\ll N/\log N \asymp b$ with an absolute implied constant, and the constant is $\leq \frac{1}{2}$ once $N\|\beta\|$ exceeds an absolute threshold; below that threshold, i.e. for $\|\beta\| \asymp 1/N$, the integral is $(1+o(1))b \cdot \widehat{\chi}(\beta N)$ with $\widehat{\chi}$ the Fourier transform of the indicator of $[0,1]$, whose real part is $\leq \frac{1}{2}$ off the principal arc. For $q=2$ the sign is reversed, so $\operatorname{Re} S_A \leq 0$ near $\theta = \frac{1}{2}$ — the helpful direction. No Siegel–Walfisz is used. $\square$

Lemma 4.12 (the small-q ring)**.** *Let $3 \leq q \leq (\log N)^{A_0}$, $(a,q)=1$, and $\theta = a/q+\beta$ with $|\beta| \leq 1/(q\tau)$, $\tau = N(\log N)^{-A_0}$. Then*

$$S_A(\theta) \;=\; \frac{\mu(q)}{\varphi(q)}\,W(\beta) \;+\; O\big(Ne^{-c\sqrt{\log N}}\big), \qquad W(\beta) = \int_2^N e(t\beta)\,\frac{dt}{\log t},$$

and $|W(\beta)| \leq \operatorname{li}(N) = (1+o(1))b$. Consequently $\operatorname{Re} S_A(\theta) \leq (\frac{1}{2}+o(1))b$.

Proof. Sort the primes $p \leq N$ by their residue class. The $\omega(q) = O(\log q)$ primes dividing q contribute $O(\log q) = o(b)$, so

$$S_A(\theta) \;=\; \sum_{\substack{r \bmod q\\(r,q)=1}} e\Big(\frac{ra}{q}\Big) \sum_{\substack{p\leq N\\ p\equiv r\,(q)}} e(p\beta) \;+\; O(\log q).$$

By Siegel–Walfisz, $\pi(t;q,r) = \operatorname{li}(t)/\varphi(q) + O(te^{-c\sqrt{\log t}})$ uniformly for $q \leq (\log N)^{A_0}$; partial summation against $e(t\beta)$ turns this into

$$\sum_{\substack{p\leq N\\ p\equiv r\,(q)}} e(p\beta) \;=\; \frac{W(\beta)}{\varphi(q)} \;+\; O\big(Ne^{-c\sqrt{\log N}}(1+|\beta|N)\big),$$

and $|\beta|N \leq (\log N)^{A_0}/q$, so the error is $O(Ne^{-c\sqrt{\log N}}(\log N)^{A_0}) = O(Ne^{-c'\sqrt{\log N}}) = o(b)$: a fixed power of $\log N$ cannot compete with $e^{-c\sqrt{\log N}}$. Summing over r and using the Ramanujan sum $\sum_{(r,q)=1} e(ra/q) = c_q(a) = \mu(q)$ for $(a,q)=1$ gives the display. Finally $\mu(q)/\varphi(q)$ is real and $\varphi(q) \geq 2$ for $q \geq 3$, so $\operatorname{Re} S_A(\theta) \leq (|\mu(q)|/\varphi(q))b(1+o(1)) \leq (\frac{1}{2}+o(1))b$; Lemma 4.17 says where the $\frac{1}{2}$ is attained. $\square$

Remark 4.13 (where the ineffectivity is)*.* The Siegel–Walfisz step in Lemma 4.12 is the only inexplicit estimate in §4.2 — Lemmas 4.8, 4.11, 4.17 and the Vaughan quotation in Lemma 4.10(c) are all effective — and it is the same constant that Part II isolates as C_1; see Remark 7.1.

Remark 4.14 (what would make C_1 effective). Two escapes are worth recording, because C_1 is the only inexplicit quantity in the trilogy. *Under GRH*, the Siegel–Walfisz input of Lemma 4.12 may be replaced by $\pi(t; q, r) = \mathrm{li}(t)/\varphi(q) + O(t^{1/2} \log t)$ with an absolute implied constant, uniformly for $q \le t^{1/2}$. The range Lemma 4.12 actually uses, $q \le (\log N)^{A_0}$, is far inside that, so on GRH the constant becomes effective and every constant in the trilogy is explicit. *Unconditionally*, the ineffectivity is confined to the possible existence of a Siegel zero for a single modulus $q \le (\log N)^{A_0}$: for every modulus with an explicit zero-free region the estimate is already effective. The inexplicit constant is therefore not a feature of the argument but a hostage to one hypothetical zero, and the range it is held over is only $(\log N)^{A_0}$, not N. Neither escape is used below; the theorems are stated unconditionally.

Remark 4.15 (the middle range, and why no constant enters). Inside $|\theta| < 1/(2N)$ Lemma 4.10 says nothing, and it does not have to: there $|a\theta| \le \frac{1}{2}$ for every $a \in A$, so $\sin^2(\pi a\theta) \ge 4a^2\theta^2$ and Lemma 4.8 gives the quadratic bound $|\widetilde{G}_B(\theta)| \le \exp(-4t_{\min} V_d \theta^2)$ directly. Cutting at the LCLT-natural radius $\theta_0 = (\log k)/\sqrt{K_d''}$ of §4.3, this is $\le \exp(-t_{\min}(\log k)^2)$ on the whole of $\theta_0 \le |\theta| \le 1/(2N)$ — superpolynomially below the density scale $(K_d'')^{-1/2}$, with no free constant anywhere. The crossover at $\theta = 1/(2N)$ survives as the boundary against Lemma 4.10, forced by $|a\theta| \le \frac{1}{2}$ rather than chosen; measured, the quadratic bound has already reached $e^{-0.01b}$ by $\theta N = 0.18$–0.42 across three profiles. [experimentally confirmed; `gap_r090`]

Remark 4.16 (what part (c) does *not* need). Part II must control every harmonic $n\theta$ with $n \le N_0$, and pays for it with a covering lemma and the parameter $Q_2 = (\log N)^{A_0 + A_0'}$. Here only $n = 1$ occurs, so the dichotomy in Lemma 4.10 is on the single denominator $q_1(\theta)$ and no covering lemma is needed. The quotation in (c) is therefore valid on a strictly larger set of θ than Part II's type (ii).

Lemma 4.17 (the extremiser of the additive estimate)**.** $\max\limits_{q \ge 3} \dfrac{\mu(q)}{\varphi(q)} = \dfrac{1}{2}$, *and the maximum is attained at $q = 6$ and at no other q.*

Proof. $\varphi(q) = 2$ holds exactly for $q \in \{3, 4, 6\}$, where μ takes the values -1, 0, $+1$; so among these the maximum is $\mu(6)/\varphi(6) = 1/2$. Every other $q \ge 3$ has $\varphi(q) \ge 4$ and hence $\mu(q)/\varphi(q) \le 1/4 < 1/2$. □

Remark 4.18 (the same $q = 6$, twice). By Lemma 4.10(b) the worst point of the additive estimate is therefore $\theta = 1/6$, with $\mathrm{Re}\, S_A/b \to \frac{1}{2} = \cos 60°$. Part II's multiplicative estimate has its worst point at the same q: $\sup_q M(q) = M(6) = \sqrt{3}/2 = \cos 30°$. **Same extremiser, same cause, two functionals** — both say that the odd primes lie in $\pm 1 \pmod 6$ and nowhere else, read once through $\prod |\cos|$ and once through $\sum \cos$. Measured on the odd primes below 10^7, $\mathrm{Re}\, S_A(1/6)/b = 0.499998$ against the predicted $1/2$, and $q = 6$ is the maximiser over $2 \le q \le 16$ at each of $N = 10^5, 10^6, 10^7$. [experimentally confirmed; `eta_r090`]

Remark 4.19 (the honest form of the tilt constant). $t_{\min} = t_{\min}(\rho)$ is positive for every fixed $\rho \in (0, 1)$ and bounded below on compact subsets, which is all Lemma 4.8 needs. It is *not* governed by the approximation $sN \approx 6x$: that relation is the small-x asymptote only, and the measured ratio $sN/6x$ is 1.02–1.13 at $x = 0.10$ but 1.22–1.39 at $x = 0.30$. Measured values of $t_{\min}$: 0.91, 0.89, 0.91 at $x = 0.10$ and 0.36, 0.28, 0.34 at $x = 0.30$ for the odd numbers, squares and primes. [experimentally confirmed; `tilt_r088`, `kappa_r088`]

4.3 The tilted local limit theorem, and the assembly

Write $K_d'' = \sum_{a \in B_d} a^2 p_a (1 - p_a)$ and $\mu_d(s) = \sum_{a \in B_d} a\, p_a$ for the variance and mean of B_d under P_s, and $K_d^{(j)}$ for the higher cumulants.

Proposition 4.20 (tilted LCLT, uniform over the window)**.** *[R2 and R3 proved (they quote Lemmas 4.8 and 4.10); R1 is the classical local limit computation with p_a-weights, written here to the level of its cumulant budget] For every layer $d \le D(k)$ and every m with $|m-\mu_d(s)| \le W(k)$,*

$$P_s[\textstyle\sum_{B_d} = m] = \frac{1}{\sqrt{2\pi K_d''}} \exp\Big(-\frac{(m-\mu_d)^2}{2K_d''}\Big)(1+O(\varepsilon)), \qquad \varepsilon = \frac{|K_d'''|}{(K_d'')^{3/2}} + \frac{K_d^{(4)}}{(K_d'')^2} + \frac{W(k)\,|K_d'''|}{(K_d'')^2},$$

uniformly in d and m.

Proof sketch. Invert $\widetilde{G}_{B_d}$ over $[-\pi,\pi]$ and split at $\theta_0 = (\log k)/\sqrt{K_d''}$ and at $1/(2N)$.

R1, $|\theta| \le \theta_0$. Expand $\log \widetilde{G}_{B_d}(\theta) = i\mu_d\theta - \frac{1}{2}K_d''\theta^2 - \frac{i}{6}K_d'''\theta^3 + \frac{1}{24}K_d^{(4)}\theta^4 + O(K_d^{(5)}\theta^5)$. Since $|K_d^{(j)}| \le \sum_a a^j$, on $|\theta| \le \theta_0$ every term beyond the quadratic is $o(1)$ and the standard Edgeworth argument gives the display with the stated ε; the third term contributes the $W|K'''|/(K'')^2$ piece over a window of width W.

R2, $\theta_0 \le |\theta| \le 1/(2N)$. Remark 4.15: the contribution is $\le \exp(-t_{\min}(\log k)^2)$, super-polynomially below $(K_d'')^{-1/2}$.

R3, $|\theta| \ge 1/(2N)$. Lemmas 4.8 and 4.10: $|\widetilde{G}_{B_d}| \le \exp(-t_{\min}(1-\eta)b/4)$, exponentially small, against measure $O(1)$.

Uniformity in d: B_d differs from A in $N_d \le N_{D(k)} = (\log k)^{O(1)}$ elements of size $\le 2D(k)$, so every cumulant shifts by a relative $O(s_D/S_2) \to 0$ and one ε serves all layers. The lattice is of step 1 and the parity weighting is quoted from Part II. □

Remark 4.21 (the μ_d-shift, and why the layer term is what it is). This is the step that produces Φ, so it is worth doing once in full. Since $\sum_{a\in A} a\,p_a = n$ and $p_a = \frac{1}{2} - \frac{sa}{4} + O((sa)^3)$ for the small elements,

$$\mu_d(s) \;=\; n - \sum_{a\le 2d} a\,p_a \;=\; n - \frac{\sigma_d}{2} + w + \dots, \qquad w := \frac{s\,s_d}{4},$$

so the two evaluation points of the stratification sit at

$$(n+d) - \mu_d = +\delta_d - w, \qquad (n-d-\sigma_d) - \mu_d = -\delta_d - w. \tag{2}$$

Feeding (2) into Proposition 4.20, undoing the tilt through $r_{B_d}(m) = P_s[\sum_{B_d} = m]\, Z_{B_d}(s)\, e^{sm}$, and using $sd + s\sigma_d/2 = s\delta_d$ to pair the two points:

$$r_{B_d}(n+d) + r_{B_d}(n-d-\sigma_d) = 2\,Z_{B_d} e^{sn - s\sigma_d/2}\, \frac{e^{-(\delta_d^2+w^2)/2K_d''}}{\sqrt{2\pi K_d''}} \cosh\Big(\delta_d\Big(s + \frac{w}{K_d''}\Big)\Big).$$

Dividing by $r_A(n) \sim Z_A e^{sn}/\sqrt{2\pi K''}$ and using $Z_{B_d} e^{-s\sigma_d/2}/Z_A = 2^{-N_d}\prod_{a\le 2d}\cosh(sa/2)^{-1} = 2^{-N_d} e^{-s^2 s_d/8}(1+\dots)$,

$$\frac{r_{B_d}(n+d) + r_{B_d}(n-d-\sigma_d)}{r_A(n)} = 2^{1-N_d}\, e^{-\lambda^2 s_d/8}\, \cosh(\lambda\delta_d) \cdot e^{-\delta_d^2/2K_d''}\,(1+O(\varepsilon)), \tag{3}$$

with $\lambda = s + w/K_d'' = s(1 + s_d/(4K_d''))$ — which is exactly the per-layer slope drift of $1 + O(s_d/S_2)$ measured earlier. **The first three factors of (3) are the general term of Φ in (1)**; the odd orders in λ cancel because the two points are $\pm\delta_d$ about the same centre, which is the reflection symmetry $\mathrm{lm}(n) = \mathrm{lm}(T-n)$ in analytic dress. [Lean-verified (the symmetry)]

Remark 4.22 (the one factor Φ omits). (3) carries a fourth factor, $c_d := e^{-\delta_d^2/2K''_d}$, that Φ does not. It is λ-*independent* and equal to $1 - O(\delta_d^2/V)$, so it belongs to E_k and not to the transfer function. Measured at $k = 180$: the ratio of the left side of (3) to Φ's term is c_d to six decimal places at $d = 1$ and $d = 3$ (1.000000 and 1.000003 for the odd numbers, 1.000000 and 1.000000 for the primes), and departs from 1 only at $d = D(k)$, where the layer weight is $2^{-29} \approx 2 \cdot 10^{-9}$. Carrying c_d inside Φ would move Table 1 from 0.842% to 0.814% (odd numbers, $k = 220$, $x = 0.06$), from 0.914% to 0.902% (primes) and from 1.052% to 1.048% ($\alpha = 3/2$): the factor is real, it moves the residual the right way, and it accounts for at most 3% of it. We therefore keep Φ as stated and count c_d in E_k. [experimentally confirmed; p3_r096, phig_r096]

That choice was made at one value of k, and the level of the correction is the wrong thing to decide it on; the trend is the right thing. Since $1 - c_d \asymp \delta_d^2/2V_d$ with $V \asymp k^{2\alpha+1}$, and the series is dominated by the small d where $\delta_d = O(1)$, the c_d correction is $O(k^{-(2\alpha+1)})$ while the residual is $O(1/k)$: the *share* of the residual that c_d carries should fall like $k^{-2\alpha}$. It does. Writing that share as $(R_\Phi - R_{\Phi c})/R_\Phi$ and fitting successive k:

$$1.98 \text{ (odd numbers, } \alpha = 1), \qquad 2.35 \text{ (primes)}, \qquad 2.99 \ (\alpha = \tfrac{3}{2}),$$

against the predicted 2α, i.e. 2.00, $\approx$2.4 for $a_i \asymp i \log i$, and 3.00; for the squares the prediction is k^{-4} and the correction is already at $1 - c_1 = 8.9 \cdot 10^{-10}$, so its share is numerical noise and changes sign — the prediction taken to its limit rather than a counterexample. Concretely the share for the odd numbers runs $10.74\%, 4.93\%, 3.33\%, 2.40\%, 1.81\%$ at $k = 120, 180, 220, 260, 300$. [experimentally confirmed, four profiles, $k \leq 300$; e8_r103] Keeping Φ as printed is therefore not a convenience at the k we happened to measure: c_d's contribution is a vanishing fraction of a vanishing residual, so Φ is the right transfer function for every larger k and c_d belongs to E_k permanently. The fork is closed.

[experimentally confirmed; p3_r096] The verification behind Proposition 4.20 and (2), at $k = 180$, two profiles, $x = 0.10$ and $x = 0.30$, layers $d = 1, 3, D(k)$: the μ_d expression is exact to a relative $6 \cdot 10^{-5}$ or better; the two offsets of (2) to $7.6 \cdot 10^{-4}$ of δ_d or better; the R1 formula against the exact r_{B_d} over the window to $2.7 \cdot 10^{-3}$–$3.7 \cdot 10^{-2}$, in every case between 4% and 14% of the predicted ε; the parity oscillation of r_{B_d} over the window at most $3.7 \cdot 10^{-3}$; and ε itself grows by at most 44% from $d = 1$ to $d = D(k)$, so a single ε does serve all layers. With Lemmas 4.8 and 4.10 the tilted generating function decays like $\exp(-t_{\min}(1-\eta)b/4)$ off the principal arc — exponentially in b, which is what P5 quotes. The tilted analogue of the $M(q)$ spectrum is lost, and is not needed: the rate enters Theorem 4.1 only through $E_k = o(1)$.

Remark 4.23 (why the constants cannot be imported). What Proposition 4.20 leaves open is not a mechanism but a list of explicit Edgeworth constants in region R1, so it is natural to ask whether they can be quoted rather than derived. The natural source is the Bernoulli-part-extraction literature, which produces local limit theorems with explicit universal constants for non-identically distributed lattice summands, and $\sigma(S) = \sum a_i X_i$ is a sum of independent lattice summands. We checked, and the answer is no, for two reasons; we record them because a reader will otherwise ask the same question.

Write, following Giuliano and Weber [3], $\vartheta_X = \sum_k \mathbb{P}\{X = k\} \wedge \mathbb{P}\{X = k+1\}$ and $\Theta_n = \sum_j \vartheta_{X_j}$; their local limit theorem carries the error $C\Theta_n^{-1/2}(H_n + \Theta_n^{-1})$ plus a term $(\log \Theta_n/(\operatorname{Var} \cdot \Theta_n))^{1/2}$, with H_n a Berry–Esseen distance, and is useful exactly when $(\operatorname{Var}/\Theta_n)^{1/2}(H_n + \Theta_n^{-1}) \to 0$.

First, the hypothesis is void here. Our summands are $X_j = a_j X'_j$ with a_j odd and at least 3, so the law of X_j charges the two integers 0 and a_j and no two *adjacent* ones. Hence $\vartheta_{X_j} = 0$ exactly, for every j and every p, so $\Theta_n = 0$ and the bound reads $\leq \infty$. Their own remark names this degenerate case.

Second, the obvious repair does not help. One can block the sequence and apply the method to block sums, whose laws do charge adjacent integers — but only once a block is large enough to have more subset sums than its span, and for blocks of *large* elements that takes a long time: $\{7, 9, 11\}$ has subset sums $\{0, 7, 9, 11, 16, 18, 20, 27\}$, no two adjacent. Measured, the smallest final block with $\vartheta > 0$ has size $7, 11, 15$ and more than 20 at $k = 16, 32, 64, 128$, growing like $\log k$, so blocking buys $\Theta_n \lesssim k/\log k$. And then the applicability condition fails outright: our weights grow, so $\mathrm{Var} \asymp S_2 \asymp k^3$ while $\Theta_n \lesssim k$ and $H_n \asymp k^{-1/2}$, giving $(\mathrm{Var}/\Theta_n)^{1/2}(H_n + \Theta_n^{-1}) \asymp k^{1/2}$. Measured, that quantity runs $48.9, 101.4, 222.5, 495.1$ for the odd numbers and $94.1, 195.1, 441.0, 1034.2$ for the primes at $k = 16, 32, 64, 128$ — increasing, not tending to zero. [experimentally confirmed; `bpart_r110`]

The obstruction is the same feature that makes the present local limit theorem worth proving: the summands carry weights $a_j \asymp j$, so the variance grows like the cube of the number of summands rather than linearly, and a method calibrated to the latter has nothing to give. *The Edgeworth constants of R1 have to be derived, not imported.*

5 Universality of the rate function

[experimentally confirmed, $k = 16..56$, three reference sequences] With $\hat{I}_A(\rho) = -k^{-1}\log(\deg_A(\lfloor \rho T \rfloor)/2^k)$ and $D_k = \hat{I}_P - \hat{I}_{\mathrm{ref}}$, the difference tends to 0 for each of three reference sequences: the odd numbers, the arithmetic progression $6n + 5$, and a random odd sequence of the same range. The sign is a property of the reference, not a universal fact — it is positive against the odd numbers, negative against the random sequence, and changes sign near $k = 48$ against the progression.

Remark 5.1 (the primes sit near the random sequence). At $k = 56$ the distance to a random odd sequence is 0.0016 against 0.0146 to the arithmetic progression — an order of magnitude. The reading we propose is that the rate function does not see the arithmetic of A beyond its counting function: the primes are close to a random sequence of the same density because, for this observable, that is what they are. It is the same statement the transfer function makes locally, since Φ depends on A only through (N_d, σ_d, s_d) — the small elements, counted. This is the most direct evidence we have for the reading that the main term is universal and the arithmetic of the sequence enters only through the correction.

6 A profile for which the hypothesis fails

The hypothesis Theorem 3.1 needs is the convergence, with room, of the series in (1) — qualitatively, that $N_d \to \infty$ fast enough. Precisely:

$$\textbf{(H)} \qquad \sup_k \sum_{d \geq 1} 2^{-N_A(d)}\delta_d^2 < \infty, \qquad \text{and} \qquad W(k) := \max\{\delta_d : 2^{-N_d}\delta_d^2 \geq k^{-1}\} = N^{o(1)}.$$

The first clause makes Φ converge and makes $\Phi(\lambda) - \Gamma = O(\lambda^2)$; the second is what the error budget needs, since the extraction window must be narrow enough for Proposition 4.20 to be uniform across it. For the odd primes $N_A(d) = \pi(2d) - 1 \gg d/\log d$, so both clauses hold, with $W(k) = O((\log k)^{2+\epsilon})$.

[experimentally confirmed] It is not vacuous. For the odd-ified cubes $a_i = 2\lfloor i^3/2 \rfloor + 1$:

profile	odds	primes	squares	cubes
N_d at $d = 10$ (any k)	10	7	4	**2**
$Q(0) = \Gamma^{-1}\sum 2^{-N_d}(\delta_d^2 - s_d/4)$	20.3	50.4	916	**381 904**

Only $\{1,9\}$ ever lie below 20, at any k, so 2^{-N_d} never damps the growing δ_d^2 and the series turns over only near $d \approx 10^4$. At $k = 70$ the finite-size offset is still $1.75 \cdot 10^{-2}$ while the signal is $1.7 \cdot 10^{-11}$: nine orders of magnitude, so the profile is not merely slow but undecidable at any accessible size. The cubes are therefore not a defect of the theory but the reason (H) has to be there: without it, nothing in §4.2 or §4.3 fails, and yet the conclusion is out of reach.

6.1 Two different kinds of failure

The cubes fail *effectively* and not asymptotically, and the distinction is worth keeping. Computed directly, $\sum_d 2^{-N_d}\delta_d^2$ for the odd-ified cubes settles at $1.112 \cdot 10^7$ and does not grow with k (values at $k = 60, 100, 140, 180$: $1.112 \cdot 10^7$ from $k \approx 140$ on, against 69.5 for the odd numbers, 266.1 for $\alpha = 3/2$ and 6473 for the squares — each of them k-independent as well). [experimentally confirmed; `alpha_r092`] So (H) is not what the cubes violate; what they violate is any hope of seeing the asymptotics, the constant being four orders of magnitude larger than the odd numbers'.

A profile that fails (H) outright is available, and on the other side:

Proposition 6.1 (the $\alpha = 1$ threshold)**.** *Let $A = A_k$ consist of k distinct odd integers with $a_i \asymp c\, i^\alpha$.*

(i) If $\alpha < 1$ then (H) *fails: $\sum_{d \geq 1} 2^{-N_d}\delta_d^2 \to \infty$.*

(ii) If $\alpha \geq 1$ then (H) *holds, with a bound depending on α and c alone.*

Proof. (i) The a_i are distinct odd integers, so $a_k \geq 2k - 1$; with $a_k \asymp ck^\alpha$ this forces $c \gg k^{1-\alpha}$, so $a_1 \asymp c \to \infty$. For $d < a_1/2$ no a_i is $\leq 2d$, hence $N_d = 0$, $\sigma_d = 0$, $\delta_d = d$, and

$$\sum_{d \geq 1} 2^{-N_d}\delta_d^2 \;\geq\; \sum_{d < a_1/2} 2d^2 \;\asymp\; a_1^3 \;\asymp\; k^{3(1-\alpha)} \;\longrightarrow\; \infty.$$

(ii) For $\alpha \geq 1$ one has $a_1 = O(1)$ and $N_d = \#\{i : a_i \leq 2d\} \gg d^{1/\alpha}$, while $\delta_d = d + \sigma_d/2 \ll d^{1+1/\alpha}$, so the general term is $\ll 2^{-c' d^{1/\alpha}} d^{2+2/\alpha}$, which is summable: an exponential in a positive power of d beats a polynomial. The bound involves α and c and not k. $\square$

Both halves are visible in the computation. For $\alpha = 1/2$ the series reads 4923, 10245, 16608, 24464 at $k = 60, 100, 140, 180$, growing like $k^{3/2}$ — which is $a_1^3 = k^{3(1-\alpha)}$ with $\alpha = 1/2$, exactly as (i) predicts. For $\alpha \geq 1$ it is flat in k to every digit printed:

profile	odds	$a_i \asymp i^{3/2}$	squares	cubes	$\alpha = 1/2$
α	1	3/2	2	3	1/2
$\sum_d 2^{-N_d}\delta_d^2$	69.5	266.07	6473.4	$1.112 \cdot 10^7$	*grows*

[experimentally confirmed, $k = 60, \ldots, 180$; `alpha_r092`] Proposition 6.1 is about clean powers, and the primes do not route through it: Theorem 3.1 for the primes quotes §4.2, and the primes' (H) is checked directly where (H) is stated, from $N_d = \pi(2d) - 1 \gg d/\log d$. No reader need look for $i \log i$ in part (ii). So **(H) cuts the power profiles at exactly** $\alpha = 1$, and the cubes sit inside the admissible range but beyond the reach of computation.

7 Honest scope

In the style of Part I [1] §7.3 and Part II [2] §11, here is what rests on what.

- *Formally verified*: the combinatorial layer, inherited from Part I, for every target.
- *Proved*: the bridge of §4.2 — Lemma 4.8 in full, Lemma 4.10 in all three parts (with (c) quoted verbatim from Part II's Step 2 at $n = 1$, and the covering apparatus of that paper not needed here, see Remark 4.16), Lemma 4.17, and Proposition 6.1. The minor-arc input is m-independent, so it transfers to off-centre targets unchanged; Part II's ineffectivity through the Siegel–Walfisz constant is inherited at the single point recorded in Remark 4.13.
- *Proof skeleton, with the analytic ingredients in place*: Theorems 3.1 and 4.1. Regions R2 and R3 of Proposition 4.20 are proved — they quote the bridge — and region R1 is the classical local limit computation with p_a-weights, given here to the level of its cumulant budget rather than with every Edgeworth constant written out. That is the one gap between this paper and a complete proof, and it is a gap of exposition rather than of ingredients. Remark 4.23 records that those constants cannot be imported from the effective local-limit literature, and why: our summands have Bernoulli part exactly 0, and their weights make the variance grow like the cube of the number of summands, which is outside the regime that literature is calibrated to.
- *Experimentally confirmed, with the range stated*: rigidity (§3), the transfer function to 0.8–1.9% of the signal (Table 1), universality (§5), the profile-dependent constant $4(2\alpha + 1)/(\alpha + 1)$ verified on four profiles.
- *Not claimed*: the residual 0.8–1.9% of Table 1 is consistent with finite size and shrinks in k, but we do not claim it is zero.

Remark 7.1 (the single ineffective constant). Precisely: every estimate used here is effective except part (b) of Lemma 4.10, whose error term is the Siegel–Walfisz error, and Part II's appendix identifies the same constant C_1 as the only ineffective one in that paper, entering there at one point as well. Part I is unconditional and effective throughout. **Ineffective is not conditional**: no statement here depends on an unproved hypothesis; what is not available is a numerical threshold beyond which the asymptotics take hold.

Remark 7.2 (approaches that did not work). As in Part II we keep the dead ends, with reasons. *The* Γ_x *family*: the idea that off-centre targets are described by a family of tilted gap series indexed by x. Rejected by the data — rewriting the observed collapse in the exact tilt parameter makes it worse, not better. *The multiplicative bridge*: $|\widetilde{G}_B| \leq |G_B|^\kappa$, which would have carried every exponent of Part II across at once. Impossible for the reason in Remark 4.9, and the underlying inequality runs the other way. *The residual as a tilt-versus-slope artefact*: the constants of Remarks 2.5 and 4.7 agree to two digits on every profile, but substituting s for λ makes the residual three times worse — a shared driver, not a cause. *A half-integer exponent* reported in an early draft, which was an artefact of the Gaussian slope (Remark 2.3). *Restating* (H) *so that the cubes violate it*: the series is bounded for the cubes and unbounded for $\alpha < 1$, and the hypothesis follows the computation rather than the narrative.

Data availability

Every number in this paper is produced by a script that writes a log beside itself, as in Parts I and II [1, 2]. The scripts named in the [status] tags — `table1_r086`, `kappa_r088`, `tilt_r088`, `sres_r088`, `arc_r088`, `eta_r090`, `gap_r090`, `saddle_r092`, `alpha_r092`, `t1row_r094`, `p3_r096`, `phig_r096`, `e8_r103`, `bpart_r110` — and their logs are part of the public artefact, together with the Lean development, at `https://github.com/mathlab0911/arithmetic-landscapes`.

References

[1] K. Amauchi, *Arithmetic landscapes I: the gap series.* Manuscript, 2026.

[2] K. Amauchi, *Arithmetic landscapes II: asymptotic flatness of subset-sum landscapes of primes.* Manuscript, 2026.

[3] R. Giuliano and M. Weber, *Approximate local limit theorems with effective rate and application to random walks in random scenery.* Bernoulli **23** (2017), 3268–3310; arXiv:1412.3980. Cited only for the definition and the applicability condition quoted in Remark 4.23.